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ORIGINAL ARTICLE

Identification and fine-mapping of *QYrAS286-2BL* conferring adult-plant resistance to stripe rust in cultivated emmer wheat

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Abstract

Key message **A novel major adult-plant stripe rust resistance QTL derived from cultivated emmer wheat was mapped to a 123.6-kb region on wheat chromosome 2BL.**

Abstract Stripe rust, caused by the fungal pathogen *Puccinia striiformis* f. sp. *tritici* (*Pst*), is one of the most devastating diseases of wheat. Identification of new sources of resistance and their utilization in breeding programs is the effectively control strategy. The objective of this study was to identify and genetically characterize the stripe rust resistance derived from the cultivated emmer accession AS286. A recombinant inbred line population, developed from a cross between the susceptible durum wheat line langdon and AS286, was genotyped using the Wheat55K single nucleotide polymorphism array and evaluated in field conditions with a mixture of the prevalent Chinese *Pst* races (CYR32, CYR33, CYR34, Zhong4, and HY46) and in growth chamber with race CYR34. Three QTLs conferring resistance were mapped on chromosomes 1BS, 2BL, and 5BL, respectively. The *QYrAS286-1BS* and *QYrAS286-2BL* were stable with major effects, explaining 12.91% to 18.82% and 11.31% to 31.43% of phenotypic variation, respectively. *QYrAS286-5BL* was only detected based on growth chamber seedling data. RILs harboring both *QYrAS286-1BS* and *QYrAS286-2BL* showed high levels of stripe rust resistance equal to the parent AS286. The *QYrAS286-2BL* was only detected at the adult-plant stage, which is different from previously named *Yr* genes and inherited as a single gene. It was further mapped to a 123.6-kb region using KASP markers derived from SNPs identified by bulked segregant RNA sequencing (BSR-Seq). The identified loci enrich our stripe rust resistance gene pool, and the flanking markers developed here could be useful in marker-assisted selection for incorporating *QYrAS286-2BL* into wheat cultivars.

Introduction

Wheat (*Triticum aestivum* L.) is the most widely cultivated and important cereal crops in the world, providing staple food for approximately 35% of the world's population (Ma

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et al. 2020). The ever-increasing world population will continue to add to wheat demand. However, wheat production encounters several constraints, including various pathogen diseases, such as rust diseases, powdery mildew, and *Fusarium* head blight (Dean et al. 2012). Stripe rust (*Puccinia striiformis* f. sp. *tritici*, *Pst*) is one of the most destructive fungal diseases of wheat globally, especially in cooler and moister environments (Chen 2005; Wellings 2011). Deploying resistant cultivars is the most economical and sustainable strategy to control this disease.

Two major types of resistance including all-stage resistance (ASR) or adult-plant resistance (APR) have been characterized and widely used in stripe rust control. All-stage resistance is mostly race-specific and can be observed at seedling stage, which is effective at all plant growth stages. Cultivars with only ASR often become susceptible with a few years of release due to the emergence of new virulent races (Chen and Kang 2017). In contrast, some APR are non-race-specific and mainly effective at the late stages of plant growth. The APR often confers a partial level of resistance but is more durable and is effective against a wider spectrum of pathogen races than ASR (Fu et al. 2009; Krattinger et al. 2009).

To date, stripe rust resistance genes *Yr1-Yr86* and many provisionally designated genes and alleles have been identified in wheat and its relatives (McIntosh et al. 2021; Xiao et al. 2022; Zhu et al. 2023). The majority of the cataloged genes confer ASR, which are no longer effective (Chen and Kang 2017; McIntosh et al. 2018). Some APR genes such as *Yr36* (Fu et al. 2009), *Yr52* (Ren et al. 2012), and *Yr59* (Zhou et al. 2014) show temperature-dependent resistance, which were usually activated at high temperature. Although a single APR gene cannot provide adequate resistance, pyramiding of APR genes can achieve a desirable resistance for wheat production (Wang et al. 2023). Therefore, identifying novel resistance genes and dissecting the underlying genetic basis of durable resistance in wheat and its relatives are important for maintaining sustainable resistance to wheat stripe rust.

Cultivated emmer wheat (*T. turgidum* ssp. *dicoccum*, $2n=4x=28$, AABB), domesticated from wild emmer wheat (*T. turgidum* ssp. *dicoccoides*, $2n=4x=28$, AABB), possesses valuable traits of resistance to diseases and tolerance to abiotic stresses and is increasingly used as a reservoir of useful genes in wheat breeding (Feng et al. 2018; Lhamo et al. 2023; Mohler et al. 2013; Terletskaya et al. 2020; Zaharieva et al. 2010). Several disease resistance genes have been previously reported from cultivated emmer wheat, such as *Pm49* (Piarulli et al. 2012) and *Pm50* (Mohler et al. 2013) for powdery mildew resistance, *Sr883-2B* and *Sr883-6A* for stem rust resistance (Sharma et al. 2019). Liu et al. (2017b) evaluated a diverse panel of 176 cultivated emmer wheat accessions against stripe rust and highlighted the presence of rich source of resistance loci that can be used for wheat improvement.

AS286 is a cultivated emmer wheat accession kept in the Genbank of Triticeae Research Institute of Sichuan Agriculture University, which originally provided by Jardin Botanique de Bordeaux, France. This line has shown high resistance to stripe rust for more than 30 years in Sichuan province, where a stripe rust epidemic has frequently occurred (Zeng and Luo 2006). The objectives of the present study were to: (1) study the genetic basis of stripe rust resistance in a *T. durum* cv. Langdon (LDN) × *T. dicoccum* AS286 recombinant inbred line (RIL) population; (2) identify and map gene(s) conferring resistance to stripe rust in AS286 using the Wheat55K SNP array and BSR-Seq; (3) construct linkage map and develop molecular markers tightly linked to *QYrAS286-2BL*; (4) narrow the physical region of *QYrAS286-2BL* and analysis candidate genes.

Materials and methods

Plant material

The mapping population comprised 102 F_8 RILs and was developed from a cross between AS286 (AS is the code of Triticeae Research Institute of Sichuan Agriculture University) and the susceptible LDN by single-seed descent. One F_8 individual plant (RIL-135) contained the target region of *QYrAS286-2BL* from AS286 and without other detect QTLs according to the QTL mapping data was selected for backcrossing to LDN to generate 215 LDN × RIL-135 F_2 plants, which were used for fine-mapping of *QYrAS286-2BL*. The susceptible common wheat line Avocet S was also used as control.

Seedling stripe rust resistance tests

Seedling tests for stripe rust resistance response were carried out under controlled growth chamber conditions as described by He et al. (2020). The 102 RILs and parents were inoculated with the Chinese prevalent *Pst* race CYR34 for seedling resistance at one-leaf stage. The RIL-135 was further tested with CRY34 and a mixture of the Chinese prevalent races (CYR32, CYR33, CYR34, Zhong4, and HY46). The LDN was used as susceptible control. Infection types (ITs) were recorded 14–21 days after inoculation using a 0–9 scale as described by Line and Qayoum (1992); the most advanced scores were used for subsequent analysis. Plants with ITs of 0–2 were recorded as highly resistant, 3–4 as quantitative resistance, 5–6 as intermediate, and 7–9 as susceptible, respectively.

Field resistance evaluation

The 102 RILs and parents were also evaluated in the field for stripe rust severity over three growing seasons (2020–2021, 2021–2022, and 2022–2023) at Wenjiang (WJ2021, WJ2022, and WJ2023), Pixian (PX2022 and PX2023), Chongzhou (CZ2023), and Shuangliu (SL2023). Wheat seeds were sown in a randomized complete block design with two replicates. Each replicate had 15 individuals in a 1.5-m row, with 30 cm between rows. The F_2 individuals of LDN × RIL-135 were tested for stripe rust resistance at the experimental field of the Triticeae Research Institute, Sichuan Agricultural University, Wenjiang. The susceptible line Avocet S was planted around the field as spreaders. All trials were inoculated with a urediniospore mixture of the Chinese prevalent races (CYR32, CYR33, CYR34, Zhong4, and HY46). The stripe rust responses were determined by infection type (IT) and disease severity (DS). IT data were recorded using a 0 to 9 scale (Line and Qayoum 1992), and disease severity was estimated based on the modified Cobb scale (Peterson et al. 1948). IT and DS were recorded three times at 7-day intervals when the flag leaves of Avocet S were fully infected. IT and DS of non-segregating lines were recorded as single values, and those of segregation lines were recorded as two or more values that were later averaged for each line. IT data and the maximum DS (MDS) were used in subsequent analysis.

Phenotypic analysis

Analysis of variance (ANOVA) and Pearson's correlation coefficients were conducted based on the mean IT and MDS values of RILs across each environment using the ANOVA function in QTL IciMapping V4.2 with the default parameters. The broad-sense heritability (h^2) of resistance across environment was estimated according to Smith et al. (1998).

SNP genotyping

Genomic DNA was extracted from the fresh leaves using Hi-DNA Plant Kit (Tiangen Biotech Co. Ltd, Beijing, China) according to the manufacturer's instructions. DNA quality and quantity were assessed using 0.8% agarose gel electrophoresis and NanoDrop 2000 (Thermo Fisher Scientific, Waltham, MA). The DNA of parents and RILs was genotyped using Wheat55K SNP array (Beijing, http://www.cgmb.com.cn/). The procedure for SNP genotype calling and allele clustering was performed as described by Wang et al. (2022).

In addition, equal amounts of RNA from 30 resistant and 30 susceptible plants at adult-plant stage from segregating F_2 population of LDN × AS286 were chosen for pooling and bulked segregant RNA sequencing (BSR-Seq) analysis. The RNA samples were sequenced using Illumina HiSeq platform at the Beijing Novogene Technology (Beijing, China) (https:// www. novogene. com/). BSR-Seq data were processed following the pipeline as described by Xie et al. (2020). In brief, the raw data were polished using software Trimmomatic v0.36 (Bolger et al. 2014) to obtain high-quality clean data. The clean reads of AS286, LDN, resistant and susceptible bulks were aligned to the reference genome of wild emmer wheat (Avni et al. 2017) for subsequent SNP calling. SNP variants with an allele frequency difference (AFD) ≥ 0.8 and p values of Fisher's exact test (FET) < $1e^{-8}$ were identified and further selected to develop SNP markers.

Linkage map construction and QTL analysis

Linkage map construction and QTL mapping were performed using IciMapping 4.2 (https://www.isbreeding.net/software/), and JoinMap 4.0. SNP markers were binned using the "BIN" function with the parameters: missing rates = 20% and distortion value = 0.01. Only one marker from each bin was randomly selected to construct the genetic map with the "MAP" function. The genetic distances were estimated using the Kosambi mapping function (Kosambi 1944). The phenotypic data (IT and MDS) were used to detect QTL using Inclusive Composite Interval Mapping with the additive tool (ICIM-ADD) in the biparental population (BIP) module of IciMapping v4.2. The minimal logarithm of odds (LOD) score to declare the presence of a QTL was set at 2.5. The phenotypic variances explained by single QTL and the additive effects at the LOD peaks were also estimated. QTLs with more than 10% of explained phenotypic variation were considered as major ones, and those sharing common flanking markers were treated as a single one.

Kompetitive allele-specific PCR (KASP) assays

The SNPs located in the interval of *QYrAS286-2BL* were selected to design KASP primers and tested polymorphisms on the parental lines. Polymorphic markers were genotyped on the entire RIL population and F_2 population of LDN × RIL-135 segregating for *QYrAS286-2BL*. KASP assay was conducted in a 10.0 µl reaction volume containing 5.0 µl of master mix (Biosearch Technologies), 1.4 µl of primer mix, 50.0 ng of genomic DNA, and 2.6 µl of ddH_2O. The CFX96 Real-Time PCR detection system (Bio-Rad, Hercules, CA) was used for amplification and fluorescence end-point reading as described by Wang et al. (2022).

Candidate gene analysis

The sequences of all closely linked SNP markers were used to blast against the wild emmer wheat (Zavitan reference genome V2.0) and durum wheat (Svevo) reference

genomes to obtain their physical positions. Gene annotations within the physical positions were retrieved from the online databases (http://202.194.139.32/). Seven sets of primers specific to chromosome 2B were designed based on the annotated gene sequences of Svevo to amplify candidate genes of *QYrAS286-2BL* in AS286 and LDN (Table S1). PCR products were sent to commercial company for gene cloning and sequencing. The obtained genomic sequences were aligned to characterize sequence polymorphism.

Quantitative real-time PCR (qRT-PCR) was performed to preliminary measure the expression profiles of *QYrAS286-2BL* candidates in resistant parent AS286 upon *Pst* race CYR34 inoculation. Leaves of AS286 were sampled at 0, 1, 3, and 5 days post-inoculation (dpi). Leaf RNA was extracted using TRIzol reagent (Tiangen, Beijing, China), and first-strand cDNA was synthesized using the TaKaRa PrimeScriptTMRT Reagent Kit (Takara, Dalian, China). The qRT-PCR was performed on a Bio-Rad CFX96 Real-Time PCR detection System (Bio-Rad, USA) using SYBR qPCR Master Mix kit (Q711, Vazyme, Nanjing, China) (TaKaRa, Kyoto). The reaction mixture contained: 5 µl 2 × SYBR Green mix, 2 µl diluted cDNA, 300 nM of each primer, in a total volume of 10 µl. The PCR program was as follows: 95 ℃ for 30 s; 40 cycles of 95 °C for 10 s and 58 °C for 30 s. Transcript levels were quantified using the $2^{-\Delta\Delta CT}$ method (Livak and Schmittgen 2001). The *TaGAPDH* was used as an internal control. Three biological replicates were used for each data point. Details of primers used for qRT-PCR are listed in Table S2.

Results

Phenotypic evaluation

AS286 exhibited partial resistance (IT = 3–4) to CYR34 at the growth chamber seedling test and high level of resistance (IT = 0–1) to mixed *Pst* races at the adult-plant stage in all field environments, whereas LDN was highly susceptible (IT = 8–9) (Fig. 1a). The IT values for the RILs ranged from 1 to 9 (mean IT = 3.11), and the MDS scores ranged from 1 to 95% (mean MDS = 22.33%). Both IT (Fig. 1b) and MDS data (Fig. 1c) for the RILs displayed continuous distribution, indicating that the resistance in AS286 is quantitative in nature. Pearson's correlation coefficients of pairwise comparisons for IT and MDS ranged from 0.48 to 0.90 and 0.51–0.79 ($p < 0.001$) (Table 1), respectively. Estimates of h^2 for IT and MDS were 0.92 and 0.86, respectively (Table 2). The ANOVA showed significant phenotypic variation ($p < 0.001$) in both IT and MDS among genotypes, environments, and genotype × environment interactions (Table 2). These results indicate that large effect stripe rust resistance loci may be present in the RIL population.

Linkage map construction

In total, 13,063 out of 13,918 SNPs in the poly high-resolution group with minor allele frequencies ≥ 0.3 were selected for linkage analysis and map construction. Of these markers, 11,580 were classified into 1,056 bins. The linkage analysis results revealed that 1,030 bin markers were mapped to 17 linkage groups, spanning a total genetic distance of 2,006.12 centimorgans (cM) for all 14

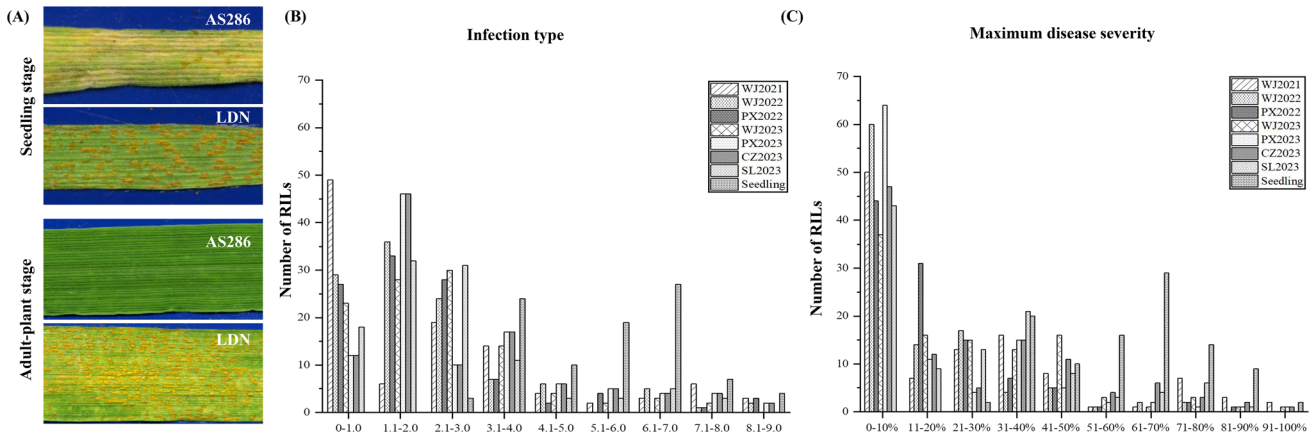

Fig. 1 Stripe rust resistance phenotype in AS286 and frequency distribution of mean infection types (IT) and maximum disease severity (MDS) for the RILs evaluated in field environments. **a** Stripe rust responses of LDN and AS286 to *Pst* race CYR34 at seedling stage and to the mixture Chinese *Pst* races at the adult-plant stage. **b** IT for the RILs was evaluated in Wenjiang, Chongzhou, Pixian, and Shuangliu during 2020–2023. **c** MDS for the RILs was evaluated in Wenjiang, Chongzhou, Pixian, and Shuangliu during 2020–2023 (colour figure online)

Table 1 Correlation coefficients (*r*) of infection type (IT) and maximum disease severity (MDS) values of the LDN × AS286 RIL population in the seven field environments

Environment	*r* value based on MDS(IT)						
	2021WJ	2022WJ	2022PX	2023WJ	2023PX	2023CZ	2023SL
2021WJ	1						
2022WJ	0.68 (0.66)**	1					
2022PX	0.53 (0.57)**	0.79 (0.65)**	1				
2023WJ	0.72 (0.67)**	0.63 (0.60)**	0.58 (0.48)**	1			
2023PX	0.53 (0.50)**	0.67 (0.60)**	0.61 (0.54)**	0.62 (0.49)**	1		
2023CZ	0.54 (0.49)**	0.62 (0.59)**	0.51 (0.54)**	0.51 (0.49)**	0.70 (0.90)**	1	
2023SL	0.71 (0.68)**	0.63 (0.59)**	0.75 (0.48)**	0.75 (0.71)**	0.63 (0.54)**	0.60 (0.54)**	1

**The *r* value significant at $p = 0.001$

Table 2 Analysis of variance (ANOVA) for infection type (IT) and maximum disease severity (MDS) scores for the LDN × AS286 RIL population

Source of variation	IT			*p* value	MDS			*p* value
	df	Mean square	*F* value		*df*	Mean square	*F* value	
RILs	105	45.01	176.50	< 0.001	105	0.36	71.79	< 0.001
Replicates	15	1.06	3.92	< 0.001	9	0.002	0.44	< 0.001
Environments	6	24.21	94.94	< 0.001	2	0.55	111.93	< 0.001
RILs × environments	630	3.82	14.98	< 0.001	210	0.05	11.16	< 0.001
Error	1890	0.25			945	0.004		
h^{2a}	0.92				0.86			

[a]Broad-sense heritability

chromosomes (Tables S3, 4). One linkage group was constructed for each chromosome except chromosomes 1A, 6A, and 7A each represented by two. The lengths of maps ranged from 19.75 (1A-2) to 166.60 cM (2A). The number of bin markers on maps ranged from 3 (6A-2) to 104 (5A). The average distance of adjacent bin markers was in the range of between 1.40 (7B) and 11.45 cM (6A-2) (Table S3), with an average density of one bin marker per 1.95 cM.

QTL analysis

Three stripe rust resistance QTLs were detected and designated as *QYrAS286-1BS*, *QYrAS286-2BL*, and *QYrAS286-5BL*, respectively (Table 3, Fig. 2). All the resistance alleles were contributed by the resistant parent AS286.

QYrAS286-1BS was closely linked to *AX-94551507* and *AX-111050443* and detected in three field environments and in seedling test contributed 12.91–18.82% and 12.79–17.31% of the variance in IT and MDS across environments, respectively. *QYrAS286-2BL* was identified in all seven field environments, whereas not found in a seedling test. It was linked to markers *AX-111695763* and *AX-110950714,* accounted for 11.31–31.43% and 10.31–31.38% of the phenotypic variance in IT and MDS, respectively. *QYrAS286-5BL* was flanked by markers *AX-109386760* and *AX-111762596* within an 8.4-cM interval. It was only found in seedling test and explained 21.73% (IT) and 17.87% (MDS) of the phenotypic variance, respectively.

Using BSR-Seq data from F_2 population of LDN × AS286, a total of 21 SNPs were enriched in a 2-Mb genomic interval (822–824 Mb) on the long arm of chromosome 2B in the wild emmer reference genome v2.0. However, the BSR-Seq approach was unable to detect the *QYrAS286-1BS* target region.

Effects of QTL combinations

The RILs with different QTL combinations were selected using flanking markers for *QYrAS286-1BS* and *QYrAS286-2BL*. Based on the field test data, RILs without any QTL had infection scores (mean IT = 7.62; MDS = 79.64%) similar to that of the susceptible LDN (mean IT = 8.25; MDS = 85.00%) in all environments. Lines with single *QYrAS286-2BL* showed higher level of resistance than those of lines with *QYrAS286-1BS*. RILs with *QYrAS286-2BL* and *QYrAS286-1BS* combination showed high level of resistance equal to resistant parent AS286. Notably, the RILs carrying *QYrAS286-2BL* alone displayed a low infection score (IT = 1.91) that was close to the effect of 1BS and 2BL QTL combination (IT = 1.42; Fig. 3), indicating that *QYrAS286-2BL* had the highest reductions in infection scores.

Table 3 Summary of stripe rust resistance QTL detected in the LDN × AS286 RIL population in field and seedling tests

QTL	Environment	Genetic position (cM)	Left marker	Right maker	LOD[a]	PVE (%)[b]	add[c]
QYrAS286-2BL	2021WJ	158.57–160.62	AX-110950714	AX-111695763	8.19	31.43	−1.28
	2022WJ	158.57–160.62	AX-110950714	AX-111695763	4.49	18.19	−0.63
	2022PX	158.57–160.62	AX-110950714	AX-111695763	3.24	14.57	−0.45
	2023WJ	158.57–160.62	AX-110950714	AX-111695763	5.26	21.53	−0.72
	2023PX	158.57–160.62	AX-110950714	AX-111695763	5.13	21.04	−0.80
	2023CZ	158.57–160.62	AX-110950714	AX-111695763	3.34	11.31	−0.66
	2023SL	158.57–160.62	AX-110950714	AX-111695763	3.34	21.01	−0.79
QYrAS286-1BS	2022PX	0–0.51	AX-111050443	AX-94551507	3.94	12.96	−0.43
	2023PX	0–0.51	AX-111050443	AX-94551507	4.86	15.14	−0.78
	2023CZ	0–0.51	AX-111050443	AX-94551507	4.86	18.82	−0.59
	Seedling	0–0.51	AX-111050443	AX-94551507	3.94	12.91	−0.46
QYrAS286-5BL	Seedling	150.71–159.12	AX-109386760	AX-111762596	5.21	21.73	−0.75

[a]*LOD* logarithm of odds score; [b]*PVE* phenotypic variance explained; [c]*add* additive effect of resistance allele. A negative value indicates that the resistance allele is from AS286

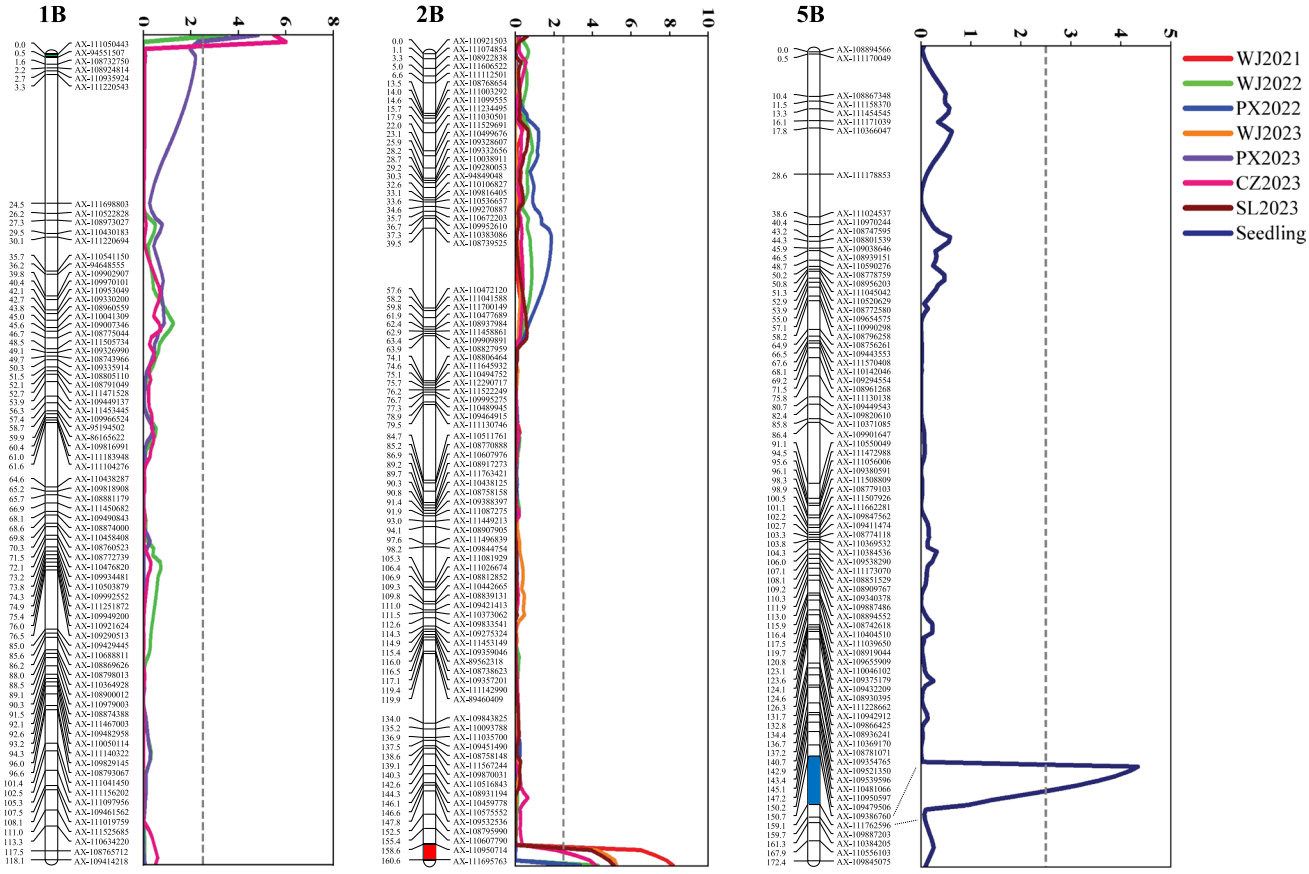

Fig. 2 Linkage maps and LOD scores obtained by inclusive composite interval mapping of QTLs for stripe rust response. Linkage maps of stripe rust resistance loci by using RIL lines of cross LDN × AS286. WJ2021, WJ2022, PX2022, WJ2023, PX2023, CZ2023, SL2023 are the ITs at Wenjiang in 2021, 2022, Pixian in 2022, Wenjiang in 2023, Pixian in 2023, Chongzhou in 2023, and Shuangliu in 2023, respectively. Seedling representsthe IT from growth chamber seedling test inoculated with CYR34. BSR-Seq analysis of F_2 individuals of LDN × AS286 with contrasting responses to *Pst* further showed the enrichment of the SNPs on chromosome 2BL (data not shown) (colour figure online)

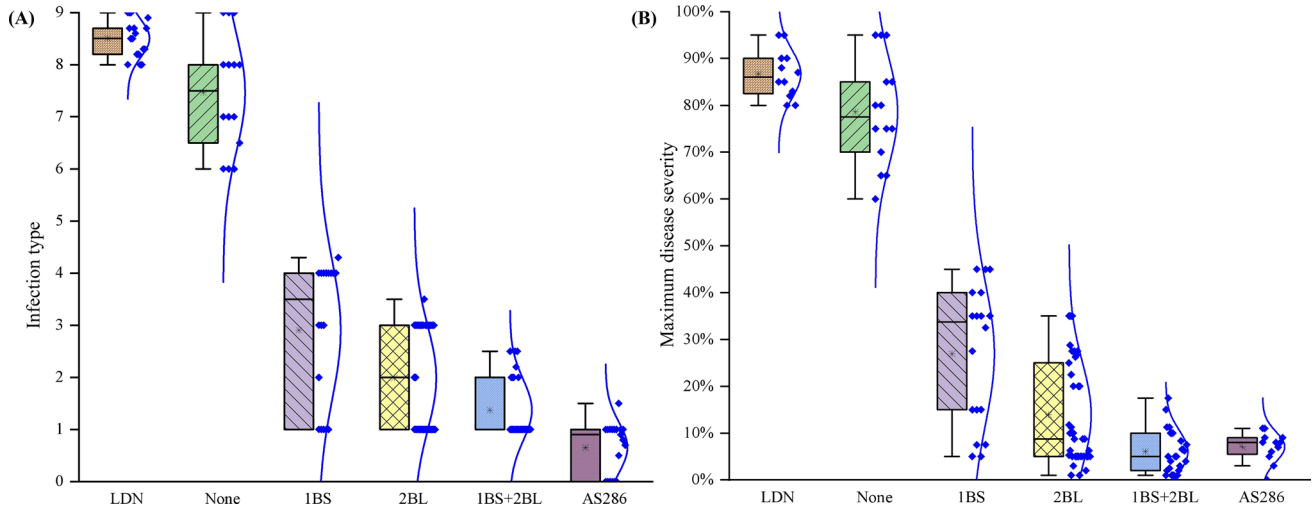

Fig. 3 Boxplot of gene interaction among *QYrAS286-1BS* and *QYrAS286-2BL* in the LDN × AS286 population. Blue dots indicate the IT (**a**) or MDS (**b**) values of individuals (colour figure online)

Genetic analysis and fine-mapping of *QYrAS286-2BL*

QYrAS286-2BL is closely linked to markers *AX-111695763* and *AX-110950714* (Table 3). Based on these markers, we selected RIL-135 which carrying homozygous *QYrAS286-2BL* region to backcross with LDN to generate segregating population of *QYrAS286-2BL*. RIL-135 was susceptible (IT = 7–8) to *Pst* race CYR34 and mixture races (CYR32, CYR33, CYR34, Zhong4, and HY46) at the seedling tests, whereas it showed high level of resistance (IT = 1–2) to the mixture *Pst* races in the field trials (Fig. 4a). F_1 plants were resistant (IT = 1) in the field. The F_2 population from LDN × RIL-135 segregated in 158 resistant (IT = 1–4) and 57 susceptible (IT = 8–9), fitting a 3R: 1S ratio ($\chi^2 = 0.109$, $p = 0.741$), indicating the dominant inheritance of *QYrAS286-2BL*.

Based on 55 K array and BSR-Seq, four candidate SNPs tightly linked to *QYrAS286-2BL* were successfully converted into KASP markers (*AX-109842666*, *KASP2B-1*, *KASP2B-2*, and *KASP2B-3*) (Table S5) and used to screen a population of 215 F_2 plants derived from LDN × RIL-135. The recombinant plants were used for fine -mapping (Fig. 4b). Finally, the *QYrAS286-2BL* was placed within a 1.2-cM genetic interval flanked by markers *KASP2B-3* and *KASP2B-1* corresponding to a 123.6-kb physical region (824,663,811–824,787,436 bp) in chromosome 2B of the Zavitan reference genome V2.0 and a 146.38-kb region (ch rUn:40,800,518–40,946,892 bp) in Svevo assembly.

Candidate genes in the *QYrAS286-1BS* and *QYrAS286-2BL* regions

The *QYrAS286-1BS* region was flanked by markers *AX-94551507* (11,947,847 bp) and *AX-111050443* (14,054,670 bp) within a 2.11-Mbp physical interval in the Zavitan reference genome assembly and a 2.04-Mbp (8,122,280–10,158,275 bp) in the durum Svevo 1BS chromosome. There were 36 and 26 predicted genes in the target physical regions in Zavitan and Svevo (Table S6). A total of 15 genes encode for protein domains that may be associated with disease resistance, such as the NB-ARC protein, protein kinases, and Leucine-rich repeat domain proteins (Table S6). The 123.6-kb and 146.38-kb interval between markers *KASP2B-1* and *KASP2B-3* of *QYrAS286-2BL* in Zavitan and Svevo had seven annotated genes (Fig. 4c, Table S7), including a leucine-rich receptor-like kinase gene, *TRIDC2BG090880*.

We further designed PCR primers to amplify these seven genes in AS286 and LDN. Four genes (*TRIDC2BG090840*, *TRIDC2BG090850*, *TRIDC2BG090870*, and *TRIDC2BG090880*) were not amplified in LDN. Therefore, we compared the genomic sequences of these genes in AS286 with reference genome sequences of Zavitan and Svevo. Sequence alignment results showed that SNP/InDels were detected in either AS286 vs. Zavitan or AS286 vs. Svevo for all seven genes (Fig. S1). Further expression analysis indicated that four genes (*TRIDC2BG090820, TRIDC2BG090850*, *TRIDC2BG090870*, and *TRIDC2BG090880*) were significantly downregulated upon the *Pst* CYR34 inoculation, whereas two genes (*TRIDC2BG090890* and *TRIDC2BG090900*) were significantly upregulated by the *Pst* CYR34 infection in AS286 (Fig. 4d). Therefore, all of these genes except *TRIDC2BG090840* represent reasonable candidates for *QYrAS286-2BL*.

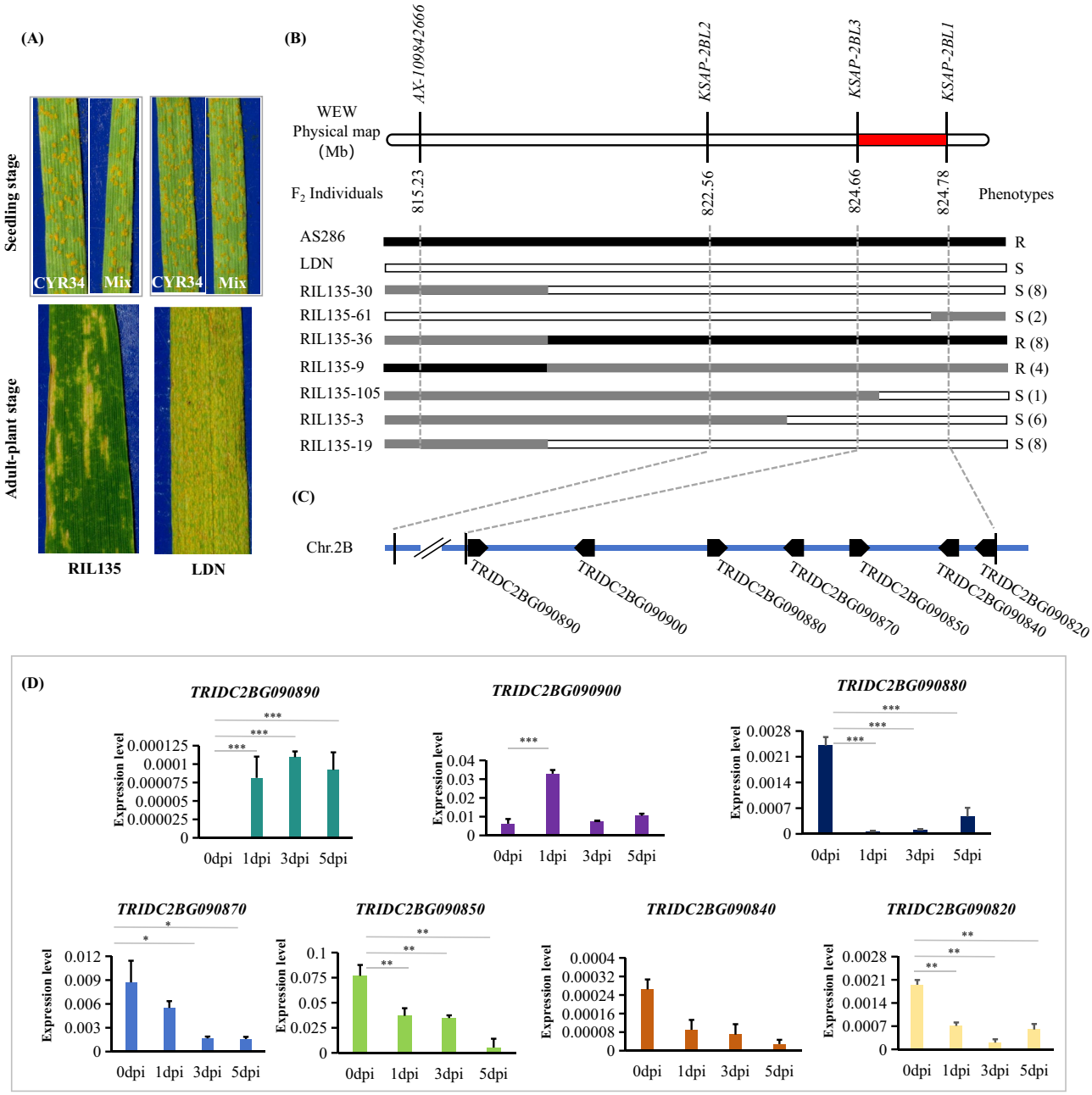

Fig. 4 Fine-mapping of the *QYrAS286-2BL* gene. A, stripe rust responses of RIL-135 and LDN to *Pst* race CYR34 (CYR34) and mixture Chinese *Pst* races (mix) at seedling stage and to the mixture races at the adult-plant stage. B, Physical positions of the development molecular markers used for fine-mapping of *QYrAS286-2BL* gene and recombinant events determined by genotypes using markers and phenotype using the F_2 plants. Black, white and gray segments represent regions homozygous for AS286, regions homozygous for LDN, and heterozygous regions, respectively. The actual number of each recombinant events is listed in the brackets. C, Annotated genes in the Zavitan genome assembly. D, Expression patterns of candidate genes of *QYr.AS286-2BL* in AS286 upon *Pst* inoculation at different time points. Asterisks indicate the level of significance, $*p<0.05$; $**p<0.01$; $***p<0.001$ (colour figure online)

Discussion

The discovery and deployment of novel resistance genes to breed disease-resistant cultivars is an effective and sustainable strategy to controlling stripe rust in wheat. Cultivated emmer wheat, one of the oldest domesticated crops in the world, is a valuable source of resistance to stripe rust (Liu et al. 2017b). Cultivated emmer accession AS286 has been highly resistant to stripe rust for over 30 years in all field experiments of Sichuan province, where a stripe rust epidemic has frequently occurred. This accession did not contain the cloned *Yr* genes, such as *Yr5*, *Yr10*, and *Yr15*

(data not shown). In the present study, three resistance QTLs in AS286 were identified through QTL analysis, including *QYrAS286-1BS* conferring ASR, *QYrAS286-2BL* for APR, and *QYrAS286-5BL* which was only detected based on growth chamber seedling data. Our results showed that the combination of these QTLs can improve the stripe rust resistance in wheat.

QYrAS286-1BS was flanked by SNP markers *AX-108732750* and *AX-94551507* within a physical interval between 11.95 Mbp and 14.05 Mbp in the Zavitan 1BS chromosome. This QTL was detected in three field environments and growth chamber seedling tests, indicating that *QYrAS286-1BS* confers ASR resistance. Several *Yr* genes including *Yr10* (Yuan et al. 2018), *Yr15* (Klymiuk et al. 2018), *YrAlp* (Lin and Chen 2007), *YrC142* (Wang et al. 2009), *YrSM139-1B* (Zhang et al. 2016), *Yr64* (Cheng et al. 2014), *Yr65* (Cheng et al. 2014), and *Yr84* (Klymiuk et al. 2022) have been previously reported on 1BS (Table S8). The physical position of *Yr10* (Yuan et al. 2018) is distal to *QYrAS286-1BS*. *Yr15* (Klymiuk et al. 2018) localizes around 83 Mbp proximal to *QYrAS286-1BS* and the *YrC142*, *YrSM139-1B*, *YrAlp*, *Yr64*, and *Yr65* genes all localize proximal to *Yr15*. The results indicate that *QYrAS286-1BS* is different to these *Yr* genes on 1BS. *Yr84*, an effective ASR gene derived from wild emmer wheat, was mapped between 9.66 Mb and 11.99 Mb (Klymiuk et al. 2022) in the Zavitan reference genome assembly, overlaps with *QYrAS286-1BS* (11.95–14.05 Mbp on Chr.1B). Liu et al. (2017b) performed a GWAS study of cultivated emmer wheat accessions and detected *QYrTtd-1BS.1* for stripe rust resistance on chromosome 1BS, which was placed close to *QYrAS286-1BS*. Thus, the concise location relationships with the three QTLs need to be further confirmed.

QYrAS286-2BL was linked to the markers *AX-110950714* and *AX-111695763* in the initial QTL analysis. Subsequent BSR-Seq analysis and fine-mapping narrowed the target region to a 123.6-Kb physical interval (824,663,811–824,787,436 bp) on chromosome arm 2BL. Several catalogued genes conferring ASR resistance to stripe rust including *Yr5* (Marchal et al. 2018), *Yr7* (Marchal et al. 2018), *Yr43* (Cheng and Chen 2010), *Yr44* (Sui et al. 2009), *Yr53* (Xu et al. 2013), and *Yr72* (Baranwal et al. 2021; Chhetri 2015) have been mapped on chromosome 2BL. Among them, *Yr5* was identified from *T. spelta* var. *album* and *Yr53* from *T. turgidum* var. durum; the remaining *Yr* genes were all characterized from common wheat cultivars. *QYrAS286-2BL* conferring APR is derived from cultivated emmer wheat. Based on the positions of the closest flanking markers, all these previously reported *Yr* genes localize on 2BL ranged from 566.98 Mb to 822.17 Mb; thus, they are different from *QYrAS286-2BL* (824.66 to 824.78 Mb). Previous studies also reported many QTLs on 2BL (Table S8, Fig. S2), such as *QYrSM155.1* (Zhou et al. 2022), *QYrqin.nwafu-2BL.1* (Zeng et al. 2019), *QYr.caas-2BL_Naxos*, *QYr.inra-2BL_CampRemy*, *QYraq.cau-2BL_Aquileja*, *QYr.caas-2BL*, *QYr.nwafu-2BL*, *QYR1* and *QYrdr.wgp-2BL_IWA6286* (Chen and Kang 2017). These QTLs resided in the interval of the catalogued *Yr* genes on 2BL that all proximal to *QYrAS286-2BL*. Therefore, *QYrAS286-2BL* may be a new APR resistance locus for stripe rust.

QYrAS286-5BL was only detected based on growth chamber seedling test, indicating that this QTL may confer race-specific resistance to CYR34, whereas ineffective to the mixture races (CYR32, CYR33, CYR34, Zhong4, and HY46) in the field. This QTL was flanked by *AX-109386760* (706.38 Mb) and *AX-111762596* (714.20 Mb) on chromosome 5BL, explained 21.73% (IT) and 17.87% (MDS) of the phenotypic variance, respectively. *Yr74* conferring ASR is the only formally documented stripe rust resistance gene mapped on 5BL (Dracatos et al. 2016). It interacted with *Yr73* to produce a unique seedling specificity resistance to A-*Pst* pts, either *Yr73* or *Yr74* alone was unable to confer resistance. Several QTLs for stripe rust resistance have been mapped on chromosome on 5BL, and most of them have minor or moderate effects (Chen and Kang 2017). These QTLs were located between 489.42- and 686.03-Mb map positions on the chromosome arm 5BL (Table S8). In the present study, *QYrAS286-5BL* was resided near the distal end of chromosome arm 5BL (706.38–714.20 Mb), which may be distinguishable from the other *Yr* genes on 5B.

In the present study, 15 candidates including seven NB-ARC, seven protein kinases, and one Leucine-rich repeat domain genes were predicted as the causal genes for *QYrAS286-1BS*. Most race-specific resistance genes identified in wheat encode nucleotide-binding domain leucine-rich-repeat (NLRs) immune receptors, which recognize pathogen effector proteins delivered into the host cytoplasm (Sanchez-Martin and Keller 2021). The kinases in plants are important signaling proteins that regulate plant cellular activities in response to biotic/abiotic stresses (Liang and Zhou 2018). Leucine-rich repeats (LRRs) play important roles in the recognition of corresponding ligands and signal transduction networks in plant defence responses (Cheng et al. 2017). Therefore, these candidates could be prioritized for further validation.

The *QYrAS286-2BL* region had seven annotated genes (*TRIDC2BG090890*, *TRIDC2BG090900*, *TRIDC2BG090880*, *TRIDC2BG090870*, *TRIDC2BG090850*, *TRIDC2BG090840*, and *TRIDC2BG090820*). Among them, *TRIDC2BG090880* encodes a leucine-rich receptor-like kinase protein. The LRR-RLK family is the largest group of plant RLKs and plays crucial roles in development and stress responses (Liu et al. 2017a). Some LRR-RLKs respond to biotic stresses, such as LRR-RLK FLS2-mediated plant resistance against bacterial pathogens (Robatzek et al. 2007), and a LRR-RLK contributed

downy mildew resistance in *Arabidopsis* (Hok et al. 2011). However, based on gene sequences and expression analyses, all seven genes except *TRIDC2BG090840* represent reasonable candidates for *QYrAS286-2BL*. Additionally, some previously cloned disease resistance genes such as *Yr36* (Fu et al. 2009) and *Pm69* (Li et al. 2023) were not present in the reference genomes. It is also possible that AS286 genome contains unique disease resistance genes for *QYrAS286-2BL*.

Some APR are controlled by non-race-specific genes that are only effective in mature plants. Although a single APR gene does not confer adequate resistance, pyramiding different type of R genes provides a promising strategy for maintaining sustainable resistance to wheat stripe rust (Huerta-Espino et al. 2020; Wang et al. 2023). The combinations of effective ASR and APR genes (*QYr.sicau-1BL* + *QYr.sicau-2AS*, *Yr46* + *Yr58*) can effectively improve stripe rust resistance (Chhetri et al. 2016; Wang et al. 2022). Our previous study showed that the APR genes *Yr28* + *Yr36* and *Yr18* + *Yr28* + *Yr36* combinations can achieve adequate resistance at the seedling stage (Wang et al. 2023). In the present study, *QYrAS286-2BL* provides a high level of APR to a mixture of the Chinese prevalent races (CYR32, CYR33, CYR34, Zhong4, and HY46), suggesting its usefulness for wheat stripe rust resistance improvement. We found that the combination of APR (*QYrAS286-2BL*) and ASR (*QYrAS286-1BS*) alleles had additive effects on stripe rust. The synthetic hexaploid wheat (SHW) line Syn-SAU-33 (AS286 × AS2407) carrying *QYrAS286-2BL* and *Yr28* showed high resistance to *Pst* (Hu et al. 2021). Therefore, the stacking of major *QYrAS286-2BL* locus with other *Yr* genes is a promising way to increase the efficiency of resistance. The linked markers of *QYrAS286-2BL* would be useful for effective implementation of the locus in wheat resistance breeding programs.

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Author contribution statement LH, LQZ, and DCL designed the experiments. YQL, YLH, YJ, QZ, YH, JSH, XJC, XC, BJ, MH, SZN, ZWY, JRZ, CJX, and LHF performed the experiments. YQL, LH, BHW, LQZ, and DCL discussed results and wrote the paper.

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Data availability All data generated or analyzed supporting the findings of the current work are available with the manuscript and its supplementary information files or are available from the corresponding authors upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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